

Functional Requirements Document (FRD)

Project Name: Construction B2B Marketplace (MVP)

Phase: 1.0 (Foundation & Data Aggregation)

Target Markets: Egypt & Saudi Arabia

Tech Stack: Next.js (Web), React Native (Mobile), NestJS (Backend), PostgreSQL (DB).

1. User Roles & Personas

Role ID	Role Name	Description	Platform Access
UR-01	Contractor (Buyer)	Procurement engineers or site managers purchasing materials.	Web (Office), Mobile (Site)
UR-02	Supplier (Seller)	Building material distributors or factories selling products.	Web (Dashboard), Mobile (alerts)
UR-03	Admin (Super User)	Platform owners managing users, disputes, and catalog.	Web (Admin Panel)

2. Modules & Functional Requirements

Module A: Authentication & Onboarding (KYB - Know Your Business)

Goal: Ensure trust by verifying business identities (Commercial Registration/Tax ID).

- **FR-A-01: Registration:** Users sign up via Email/Phone (OTP verification required).
- **FR-A-02: Company Profile:**

- Contractors must upload: Commercial Registration (CR), Tax ID, Company Logo.
- Suppliers must upload: CR, Tax ID, VAT Certificate, List of Categories served.
- **FR-A-03: Approval Workflow:** New Supplier accounts are set to Pending. Admin receives notification to manually verify documents before activation.
- **FR-A-04: Sub-Accounts:** A Company Admin can invite team members (e.g., "Site Engineer" with limited budget/permissions).

Module B: Project & Site Management

Goal: Organize orders by physical location to streamline logistics.

- **FR-B-01: Create Project:** Contractor creates a "Project" (e.g., "Riyadh Villa Compound").
- **FR-B-02: Site Location:** Integration with **Google Maps API**. User pins the exact delivery location (Lat/Long stored for logistics).
- **FR-B-03: Site Contact:** Assign a "Receiver Name" and "Phone Number" for each site.

Module C: RFQ Engine (Request for Quotation) - *The Core Feature*

Goal: Digitalize the "tendering" process. This replaces phone calls.

1. Buyer Flow (Creating RFQ)

- **FR-C-01: Manual RFQ:** Buyer selects a category (e.g., "Steel"), enters Product Name, Quantity, and Unit (Ton/Kg).
- **FR-C-02: Attachment Upload:** Buyer uploads a PDF/Excel/Image of the BOQ (Bill of Quantities).
 - *AI-Readiness Note:* The system must store this file securely linked to the structured data for future OCR training.
- **FR-C-03: Delivery Preferences:** Buyer selects "Required Delivery Date" and "Payment Terms" (Cash / Credit / Cheque).

2. Supplier Flow (Bidding)

- **FR-C-04: RFQ Feed:** Suppliers see a feed of open RFQs matching their categories.
- **FR-C-05: Submit Quote:** Supplier enters:
 - Unit Price (SAR/EGP).
 - Total Price (Auto-calculated).
 - Delivery Cost (if applicable).
 - Quote Validity (e.g., "Valid for 48 hours").
 - *Notes Field:* (e.g., "Brand is Ezz Steel").
- **FR-C-06: Partial Bidding:** Supplier can bid on specific items within a large RFQ if they don't have everything.

3. Selection Flow

- **FR-C-07: Bid Comparison:** Buyer sees a comparison table of all received quotes (Price vs. Delivery Time).
- **FR-C-08: Awarding:** Buyer clicks "Accept Quote." This converts the RFQ into an **Order**.
- **FR-C-09: Rejection Logic:** If a Buyer rejects a quote, a modal asks "Reason for rejection?" (Price too high / Delivery too late).
 - *AI-Readiness Note:* This data point is critical for the "Predictive Pricing" AI model later.

Module D: Order Management & Fulfillment

Goal: Track the lifecycle of materials moving from warehouse to site.

- **FR-D-01: Order Status Workflow:**
 - Confirmed → Processing → Out for Delivery → Delivered → Completed .
- **FR-D-02: Delivery Note (DN):** Supplier generates a digital Delivery Note PDF.
- **FR-D-03: Proof of Delivery (POD) - Mobile Only:**
 - When the truck arrives, the Site Engineer uses the App to take a photo of the materials.

- Engineer signs digitally on the screen.
- Status updates to **Delivered**.
- *Offline Capability:* Using **WatermelonDB**, this action must work without internet and sync when connection is restored.

Module E: Financials & Invoicing (Compliance)

Goal: Handle payments and legal compliance (ZATCA/ETA).

- **FR-E-01: Invoice Generation:** System auto-generates a PDF invoice upon **Delivered** status.
- **FR-E-02: Localization:**
 - **KSA:** Invoice includes QR Code (ZATCA Phase 1 standard) and Arabic text.
 - **Egypt:** Integration fields for ETA (e-invoice) standard.
- **FR-E-03: Offline Payments:** Buyers can upload a "Bank Transfer Receipt" or image of a "Cheque." Supplier manually verifies and marks order as **Paid**.
- **FR-E-04: Wallet:** A simple internal ledger showing "Total Spent" and "Outstanding Dues."

Module F: Admin Dashboard

Goal: Operational oversight.

- **FR-F-01: User Management:** Approve/Ban suppliers or contractors.
- **FR-F-02: Catalog Management:** Create Categories, Sub-categories, and Units of Measurement.
- **FR-F-03: Commission Config:** Admin can set a % commission fee per successful order (Business Model).

3. Non-Functional Requirements (NFR)

- **NFR-01: Localization:** The interface must support RTL (Right-to-Left) Arabic and LTR English fully.

- **NFR-02: Offline First:** The Mobile App (React Native) must function for Site Engineers in areas with poor signal (Caching mechanism required).
 - **NFR-03: Performance:** Dashboard load time under 2 seconds.
 - **NFR-04: Security:** All uploaded documents (Tax IDs, Contracts) must be stored in encrypted S3 buckets.
 - **NFR-05: Audit Logs:** Every action (Create, Edit, Delete, Accept) must be logged in the DB with a timestamp and UserID.
-

4. Technical Specifications Summary

- **Frontend Web:** Next.js 14 (App Router), Tailwind CSS.
 - **Mobile:** React Native (Expo recommended for speed).
 - **State Management:** Zustand (Web), TanStack Query (Data Fetching).
 - **Backend:** NestJS (Modular Monolith).
 - **Database:** PostgreSQL (v15+).
 - **ORM:** Prisma or TypeORM.
 - **Storage:** AWS S3 or Google Cloud Storage (for PDFs/Images).
-

Untitled